Review: Practical Application of diff and patch

This reading contains the code used in the instructional videos from **Practical application of diff and patch**.

Introduction

This follow-along reading is organized to match the content in the video that follows. It contains the same code shown in the next video. These code blocks will provide you with the opportunity to see how the code is written and can be used as a reference as you work through the course.

You can follow along in the reading as the instructor discusses the code or review the code after watching the video.

```
cp disk_usage.py disk_usage_original.py
cp disk_usage.py disk_usage_fixed.py
```

File with code

```
#!/usr/bin/env python3

import shutil

def check_disk_usage(disk, min_absolute, min_percent):
    """Returns True if there is enough free disk space, false otherwise."""
    du = shutil.disk_usage(disk)
    # Calculate the percentage of free space
    percent_free = 100 * du.free / du.total
    # Calculate how many free gigabytes
    gigabytes_free = du.free / 2**30
    if percent_free < min_percent or gigabytes_free < min_absolute:
        return False
    return True

# Check for at least 2 GB and 10% free
if not check_disk_usage("/", 2*2**30, 10):
    print("ERROR: Not enough disk space")
```

```
    return 1

print("Everything ok")
return 0


./disk_usage_fixed.py

#this throws an error
```

Code output:

File "./disk_usage_fixed.py", line 19

    return 1

    ^

SyntaxError: 'return' outside function

File with code

The instructor adds `import sys` at the beginning, then change return 1 to `sys.exit(1)` and `return 0` to `sys.exit(0)`.

```
#!/usr/bin/env python3

import shutil
import sys

def check_disk_usage(disk, min_absolute, min_percent):
    """Returns True if there is enough free disk space, false otherwise."""
    du = shutil.disk_usage(disk)
    # Calculate the percentage of free space
    percent_free = 100 * du.free / du.total
    # Calculate how many free gigabytes
    gigabytes_free = du.free / 2**30
    if percent_free < min_percent or gigabytes_free < min_absolute:
        return False
    return True
```

```
# Check for at least 2 GB and 10% free
if not check_disk_usage("/", 2*2**30, 10):
    print("ERROR: Not enough disk space")
    sys.exit(1)

print("Everything ok")
sys.exit(0)
```

```
./disk_usage_fixed.py
```

Code output:

```
ERROR: Not enough disk space
```

File with code

Instructor changed change `check_disk_usage("/", 2*2**30, 10)` in previous file to `check_disk_usage("/", 2, 10)`.

```
#!/usr/bin/env python3

import shutil
import sys

def check_disk_usage(disk, min_absolute, min_percent):
    """Returns True if there is enough free disk space, false otherwise."""
    du = shutil.disk_usage(disk)
    # Calculate the percentage of free space
    percent_free = 100 * du.free / du.total
    # Calculate how many free gigabytes
    gigabytes_free = du.free / 2**30
    if percent_free < min_percent or gigabytes_free < min_absolute:
        return False
    return True

# Check for at least 2 GB and 10% free
if not check_disk_usage("/", 2, 10):
    print("ERROR: Not enough disk space")
    sys.exit(1)

print("Everything ok")
sys.exit(0)
```

```
./disk_usage_fixed.py
```

Code output:

Everything ok

```
diff -u disk_usage_original.py disk_usage_fixed.py > disk_usage.diff
cat disk_usage.diff
```

Code output:

--- disk_usage_original.py 2019-06-22 15:13:38.591579963 -0700

+++ disk_usage_fixed.py 2019-06-22 15:41:35.013023839 -0700

@@ -1,6 +1,7 @@

#!/usr/bin/env python3

import shutil

+import sys

def check_disk_usage(disk, min_absolute, min_percent):

"""Returns True if there is enough free disk space, false otherwise."""

@@ -14,9 +15,9 @@

return True

Check for at least 2 GB and 10% free

-if not check_disk_usage("/", 2*2**30, 10):

+if not check_disk_usage("/", 2, 10):

    print("ERROR: Not enough disk space")

-    return 1

+   sys.exit(1)

 print("Everything ok")

-return 0

+sys.exit(0)

```
patch disk_usage.py < disk_usage.diff
patching file disk_usage.py
```

Code output:

patching file disk_usage.py

```
./disk_usage.py
```

Code output:

Everything ok